

Isolation of the fold mask, refuted

Pair counts are small but not a SATZ, isolated masks do not close $\Delta^2 \log \tau$,
and Gershgorin stays open

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Abstract

Round 395 refuted three-gap of the ν -mask and left a 2–3 dominated histogram with two F_1 pairs on FRAME-A and a sparse large-gap tail. The candidate closing object was *isolation*: gap-1 pairs source-countably rare, so the 2×2 Uvarov blocks are finitely many explicit terms and the isolated singleton chain plus a thin tail sits in the $\Delta^2 \log \tau$ box and has $\lambda_{\max} < 1$. What is proved: folded consecutive small integers are not adjacent on the fold circle (PNT-free, log monotone); wrapping $\{2, 3, 4\}$ is isolated; pair counts on named windows are machine identities; the Dirichlet envelope at angular separation 2 is still $\gg 1$. What is *refuted*: $P(N)/n_\nu \rightarrow 0$ or $P \leq$ a fixed constant (core-42 has up to 16 pairs, EXT $kz = 97$ has 22, density stays $O(0.03)$); that pairs are small- n log collisions (an atom $n = e^u \approx 211$ near the cap lands on a pair bin); that isolated masks with a thin tail imply the box ($h = 40$: 3/12 IN; the r395 8/8 at $h = 80$ is an $n_{\text{ref}} = 60$ truncation, and dies at natural depth 2/8); that isolation implies $\lambda_{\max} < 1$ (1010); that isolation plus the Dirichlet hull closes Gershgorin (gersh = 1.77). No RH claim.

Status. Lemmas 1–3 are proved (finite identities). Propositions 1–3 name the refutation. No RH claim. No L^* claim.

1 Recollection

Occupied Fejér deletes the ν -support Ξ from the discrete Bernstein–Szegő cosine grid [3, 1]. Under F_1 the clusters are singletons and adjacent pairs; τ splits into 1×1 Sherman–Morrison factors, 2×2 pair blocks, and a coupling determinant κ [3]. r395 showed the mask is not three-gap and that wrapping $\{2, 3, 4\}$ (isolated singles) sat in the box on 8/8 seeds at $h = 80$, $n_{\text{ref}} = 60$, with $\lambda_{\max}(E_{n_0}) = 0.997$, while wrapping $\{1, 2, 3\}$ (pair-rich) and random F_1 did not. The candidate: the source has only two pairs on FRAME-A, so isolation is the class and the two 2×2 terms are a finite remainder.

2 Pair counts, packing, not a smallness SATZ

Lemma 1 (folded small integers are not adjacent). *Let $L = 734$ and $x_n = \lfloor L \{\log_2 n\} \rfloor$, folded $f_n = \min(x_n, L - x_n)$. For $n = 2, \dots, 63$ no consecutive pair $(n, n + 1)$ is adjacent on the fold line (0 adjacencies). PNT-free: only monotonicity of log.*

So gap-1 on the cosine grid is *not* the small- n log-collision $n < e^{1/\delta}$.

Lemma 2 (PNT-free packing is $O(n_{\text{atom}})$). *On FRAME-A the 70 source log-atoms have 24 consecutive pairs with $\Delta u < 2D$ (D the mesh). This is a packing identity, not a smallness SATZ: $P/n_\nu \not\rightarrow 0$ from log-density alone.*

Proposition 1 (pair census). *FRAME-A $w = 9$: 2 pairs on 104 atoms (density 0.019), locations (13, 14) and (67, 68). An atom $n = e^u \approx 211$ (near the cap $X = 256$) lands on a pair bin; $n \approx 97$ sits at fold-distance 2. Atom-fold adjacencies are 18 (same-bin 9) against 2 cosine pairs — aggregation, not a Steinhaus remainder. Core-42: pair counts in $[0, 16]$ (median 4), density ≤ 0.0405 , 3/42 zero-pair windows. EXT $kz = 97$: 22 pairs on 795 atoms, density 0.0277. Dead χ sample: densities 0.040, 0.039, 0, 0.041. Thus P is not a fixed constant and P/n_ν stays $O(0.03)$, not $\rightarrow 0$ on the named range.*

3 Isolation does not imply the box

Proposition 2 (discriminator). *Wrapping $\{2, 3, 4\}$ is isolated (pairs = 0, max run = 1, F_1). At $h = 40$, $n_{\text{ref}} = 40$: 3/12 seeds IN — not a class SATZ. At $h = 80$, $n_{\text{ref}} = 60$: 8/8 IN (r395 pin). At $h = 80$, $n_{\text{ref}} = 80$: 2/8 IN. The 8/8 is a truncation census, not isolation. Denser isolated wrapping $\{2, 2, 3\}$: 0/8 IN. Sparser $\{3, 4, 5\}$: 5/8 IN. Fill, not the pair counter, moves the box. Injecting k pairs into wrapping $\{2, 3, 4\}$ at $h = 40$: $k = 1$ already 0/8 IN; $k = 10$ jump up to 3.86. The source's two pairs is not a closing threshold on this class. Punching a fat gap ($d_{\text{max}} \sim 60$): 6/8 IN against 1/8 for the thin wrap — a fat tail helps. Source $d_{\text{max}}/d_{\text{min}} = 23$ is not the kill. Wrapping $\{1, 2, 3\}$ (density 0.22) and random F_1 (density 0.42) stay OUT. On core-42 the source is 42/42 IN at JUMP and $\text{corr}(\# \text{pairs}, \text{jump}) = -0.38$: pair count does not drive the source jump.*

The singleton 1×1 chain cannot be a SATZ of the box: the isolated class itself does not sit in the box at natural depth. The finitely many 2×2 terms on the source are real, and do not explain F_ε .

4 Assist: isolation does not close Gershgorin

Lemma 3 (hull still huge). *On the $h = 40$ cosine grid ($S = 79$), $1/|\sin \Delta\theta| = 12.59$ at angular separation 2. The r394 envelope $|D_n(a)| \leq 1/|\sin(a/2)|$ does not become useful under isolation.*

Proposition 3 (Assist). *At $h = 40$, $n_0 = 16$: wrapping $\{2, 3, 4\}$ has $\lambda_{\text{max}} = 0.9968 < 1$ but Gershgorin $1.768 > 1$ and row-amplification $g_A = 6.10$. 1010 is isolated and has $\lambda_{\text{max}} = 1.405 > 1$. Two injected pairs already give $\lambda_{\text{max}} = 1.020 > 1$. Random F_1 : 2.010; wrapping $\{1, 2, 3\}$: 1.858. Two-period $c = 2/3$: $\lambda_{22} = 1.0288 > 1$ [4, 2]. Isolation is not $\lambda_{\text{max}} < 1$, and isolation plus the Dirichlet hull does not close Gershgorin.*

Named kills: wrapping $\{1, 2, 3\}$ on pair density; random F_1 on density plus fill; two-period on ρ_{AP} ; scramble seed 3 pair density 0.431; pair-counter mutant (gap-2 count 44 versus gap-1 count 2 on FRAME-A); tail- $\times 2$ mutant (fattail helps, does not kill).

5 What remains

Pair density $\sim 2\%$ on the source is a *shadow* of the r395 2–3 histogram with a sparse tail: most atoms are isolated singletons, two F_1 pairs, a thin large-gap tail. That shape is not isolation as a class, not a Steinhaus set, and not a random F_1 word. Both F_ε and Assist still hang on the histogram, now with the pair counter and the isolated synthetic class ruled out as the closing lemma.

Machine check

Standalone: $w = 9$ two pairs; folded small integers; packing; wrapping $\{2, 3, 4\}$ isolated; hull $\gg 1$. Construction: $h = 40$ not uniform IN; pair-rich OUT; 1010 $\lambda > 1$ and Gershgorin open;

two-period / scramble; injection and fattail. Discovery census: `isolation_lemma_probe.py` (`--smoke / full`), core-42, χ , EXT $kz = 97$. Exit line: ISOLATION LEMMA VERIFIED.

Claim boundary

Finite identities for folded small integers and named-window pair counts; a named refutation of isolation as a closing implication for $\Delta^2 \log \tau$ and for Assist; named periodic / random- F_1 / scramble / two-period breaks. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

- [1] S. Hamann, *Three-gap of the fold mask, refuted*, standalone note, August 2026, `rh/problem/three_gap_mask.tex`.
- [2] S. Hamann, *Sign-Schur, reduced*, standalone note, August 2026, `rh/problem/sign_schur.tex`.
- [3] S. Hamann, *The $\Delta^2 \log \tau$ -field under F_1 , reduced*, standalone note, August 2026, `rh/problem/tau_field.tex`.
- [4] S. Hamann, *Coherence assist, reduced*, standalone note, August 2026, `rh/problem/coherence_assist.tex`.